

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Prepared By: Simran(AI & Data Science Intern)

Project: Internship Project – Week 3 & Week 4

Executive Summary

This project analyzes historical Superstore sales data to improve inventory planning through sales forecasting, anomaly detection, and product demand segmentation. Multiple forecasting models were evaluated, and an interactive Streamlit dashboard was developed to support business decision-making. The system helps forecast future sales, identify unusual demand patterns, and recommend inventory strategies for improved operational efficiency.

Key Findings

- Technology generated the highest sales among all product categories.
 - Sales followed a seasonal trend with higher demand during festive and year-end periods.
 - The East region showed the strongest sales performance.
 - Data quality was good, with very few missing values and no significant duplicates.
 - XGBoost achieved the best forecasting performance (**MAE = 8,826.43, RMSE = 13,342.42**) and was selected as the final model.
-

Three-Month Sales Forecast

The XGBoost model predicts stable sales for the next three months, with Technology products and the East region expected to continue performing strongly.

Considering the model's prediction error (**MAE $\approx$ 8,826**), actual sales may vary by approximately **$\pm 9,000$** around the forecast values due to seasonal demand and business conditions. Maintaining adequate safety stock is therefore recommended.

Top Three Sales Anomalies

- **Year-end holiday season:** Sales increased because of promotional campaigns and holiday shopping.
- **Festival period:** Temporary spikes occurred because of higher customer demand.
- **Low-sales weeks:** Lower sales were likely caused by reduced customer activity or seasonal fluctuations.

Both Isolation Forest and Z-Score detected similar anomaly periods, increasing confidence in the results.

Product Demand Segmentation

K-Means clustering grouped products into four demand segments.

Segment	Recommended Stocking Strategy
High Value Products	Maintain controlled inventory and avoid overstocking.
Stable Demand Products	Maintain regular inventory replenishment.
High Volume Products	Keep higher inventory to prevent stock-outs.
Growing Demand Products	Increase inventory gradually while monitoring demand.

Business Recommendations

1. Increase inventory before festive and holiday seasons because historical sales consistently rise during these periods.
2. Maintain higher stock for Technology products since they generated the highest sales and strongest forecast.
3. Allocate additional inventory to the East region because forecasting indicates continued strong demand.

Risk and Limitation

The forecasting model is based on historical sales data. Unexpected events such as supply chain disruptions, economic changes, or major promotional campaigns may affect future sales. Regular model retraining is recommended.

Conclusion

The proposed Sales Forecasting and Demand Intelligence System combines forecasting, anomaly detection, demand segmentation, and dashboard visualization into a single decision-support solution. It enables better inventory planning, supports data-driven decisions, and helps improve operational efficiency.